

Section A: Objective Questions (22 marks)

[MCQ: +1 correct, 0 for incorrect; MSQ: +2 if all correct, -1 for incorrect]

Q1. Which theory explicitly rejects the idea of a single general intelligence factor?

- A. Spearman
- B. Sternberg
- ✓ C. Thurstone
- D. Binet

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Q2. A participant studies a list of words. During testing, they are unable to freely recall many words, but when given a recognition test (old/new), they correctly identify most previously studied items. What is the most plausible explanation for this pattern?

- A. The information was not encoded into long-term memory
- B. The memory trace has decayed over time
- ✓ C. Retrieval cues during recall were insufficient to access stored information
- D. Recognition relies entirely on procedural memory systems

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Q3. Which conditions are most likely to slow down visual search performance? (MSQ – Select ALL that apply)

- ✓ A. Conjunction search tasks
- ✓ B. Increased similarity between target and distractors
- C. Feature search with a unique feature
- ✓ D. High perceptual clutter
- E. Increasing the brightness or salience of the target

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Q4. A patient can no longer form new memories after a brain injury. However, over repeated trials, their performance on a mirror-tracing task improves, even though they insist they've never done the task before. Which explanation BEST accounts for this pattern?

- A. Episodic memory is intact but procedural memory is impaired
- ✓ B. Procedural memory is intact but episodic memory is impaired
- C. Working memory is intact but long-term memory is impaired
- D. Semantic memory is impaired but episodic memory is intact

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Q5. Which processes are most likely to enhance long-term memory encoding? (MSQ – Select ALL that apply)

- ✓ A. Elaborative rehearsal
- B. Maintenance rehearsal alone
- ✓ C. Creating multiple retrieval cues
- D. Passive exposure without attention
- ✓ E. Learning highly similar material

-1 (e incorrect)

Q6. Searching for a red object among blue objects is typically faster than searching for a red square among red circles and blue squares because:

- A. It requires top-down processing
- ✓ B. It is a conjunction search
- C. It is a feature search
- D. It involves divided attention

✗

Q7. A person studies in a noisy café and later performs better on a test taken in a similar noisy environment rather than in a quiet room. Which principle is illustrated?

- A. Distributed processing
- ✓ B. Encoding specificity
- C. Elaborative rehearsal
- D. Priming

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Q8. Which statements are **TRUE** about visual search? (MSQ – Select ALL that apply)

- ✓ A. Feature search is typically faster than conjunction search
- ✓ B. Conjunction search requires focused attention
- C. Feature search is always serial
- ✓ D. Increasing distractors affects conjunction search more
- E. Conjunction search is independent of attention

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Q9. Two patients are shown drawings of common objects.

Patient A is unable to copy the drawings and struggles to match objects based on shape.

Patient B can accurately copy the drawings and match shapes, but cannot identify or name the objects.

Which interpretation best explains their deficits?

- A. Patient A has associative agnosia, while Patient B has apperceptive agnosia
- ✓ B. Patient A has apperceptive agnosia, while Patient B has associative agnosia
- C. Both patients have impairments in semantic memory
- D. Both patients have deficits in early visual processing

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Q10. Which factors influence emotional responses to music? (MSQ – Select ALL that apply)

- ✓ A. Musical features (e.g., tempo, pitch)
- ✓ B. Listener characteristics
- ✓ C. Situational context
- D. Only lyrics determine emotional response
- ✓ E. Prior experiences and memory

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Q11. Maria is learning to drive a car. At first, she has to consciously think about every action.

After several months of practice, she can drive smoothly while having a conversation. According to Fitts and Posner's Model, she has progressed from:

- A. Associative Stage to Cognitive Stage
- ✓ B. Cognitive Stage to Autonomous Stage
- C. Autonomous Stage to Associative Stage
- D. Sensory-Motor Stage to Cognitive Stage

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Q12. Which of the following findings are consistent with early selection theories of attention?
(MSQ – Select ALL that apply)

- ☒ A. Unattended information is filtered before semantic processing
- ☒ B. Physical characteristics of unattended stimuli can be reported
- ☐ C. The meaning of unattended stimuli strongly influences behaviour
- ☐ D. Processing limits occur at a late stage
- ☒ E. Selection is based on physical features

2

Q13. In motor control, a "Closed-Loop" system is characterised by:

- ☐ A. Pre-planned movement without adjustment
- ☒ B. Use of sensory feedback
- ☐ C. Homunculus organisation
- ☐ D. Reflex responses

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Q14. The "Binding Problem" explores how the brain:

- ☐ A. Misses unexpected stimuli
- ☒ B. Combines features into a single object
- ☐ C. Chooses attention filters
- ☐ D. Ignores one visual field

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Q15. Which brain structure is primarily responsible for memory consolidation?

- ☐ A. Cerebellum
- ☐ B. Prefrontal Cortex
- ☐ C. Amygdala
- ☒ D. Hippocampus

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Q16. Participants rate previously seen statements as more true than new ones, even if told some were false. Why?

- ☐ A. Strong episodic recall
- ☒ B. Familiarity increases perceived truth
- ☐ C. Effort reduction strategy
- ☐ D. Emotional valence dominates

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Q17. A student repeatedly reads a list of random words aloud but cannot recall them a day later. Another student groups the words meaningfully and relates them to personal experiences and recalls them easily. Which combination BEST explains this difference?

- ☐ A. Maintenance rehearsal vs sensory encoding
- ☒ B. Elaborative rehearsal vs maintenance rehearsal
- ☐ C. Encoding specificity vs retrieval failure
- ☐ D. Chunking vs decay

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Section B: Short Answer Questions ($3 \times 4 = 12$)

Q1. Explain any two Gestalt principles. Illustrate each with an example.

Q2. Distinguish between Sensation and Perception. Explain the fundamental difference between the "raw data" gathered from the environment and the brain's "interpretation" of that data, including the concept of perception as the brain's "best guess".

Q3. Describe the "Primacy" and "Recency" effects in memory. Identify why we tend to remember words at the beginning and end of a list better than those in the middle, and explain what happens to the recency effect if there is a 30-second filled delay before recall.

... memory would have also occurred. Thus, eliminating some of the recent ones.

Section C: Long Answer Questions ($2 \times 8 = 16$)

Q1. A participant studies a list of related words (e.g., bed, rest, dream, tired) but later confidently recalls the word "sleep," which was not presented. Even when told that some memories may be incorrect, the participant remains confident.

Using this scenario, explain:

- The Deese-Roediger-McDermott (DRM) paradigm and how it leads to false recall
- The role of schema-driven reconstruction
- The distinction between familiarity and source memory
- Why are such errors difficult to inhibit
- One real-world implication (e.g., misinformation, eyewitness testimony)

explained above & they have higher attention span
Q2. Selective attention allows us to process relevant information while ignoring distractions, but there is debate about when selection occurs during processing.

- a) Explain the key assumptions of early and late selection theories of attention
- b) Compare and contrast the two approaches
- c) Briefly explain how intermediate models (e.g., attenuation theory) attempt to resolve this debate
- d) How does the system decide on early or late selection?